

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“Building Reusable IaC Framework with Built-in AI Agents for Harman Genesis Platform”

Submitted in partial fulfillment for the award of degree(21INT82)

**Bachelor of Engineering In
Department of Artificial Intelligence and Machine Learning**

Submitted by:

Adithya N Awati (1DS21AI001)

Conducted at



Harman International.

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**Building Reusable IaC Framework with Built-in AI Agents for Harman Genesis Platform**” carried out by **Adithya N Awati [1DS21AI001]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

**Signature of
Reporting Head**
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Assoc Director- IT,
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**Signature of Internship
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Prof. Kavya D N,
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Signature of HoD
Dr. Vindhya P Malagi
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External Viva:

Name of the Examiner

Signature with Date

1)

2)

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Adithya N Awati**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that this internship report is based on the work undertaken during my ongoing internship in the role of **Project Intern** at **Harman Connected Services, Bengaluru**. been successfully completed, in. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS21AI001

Place: Bangalore

NAME: Adithya N Awati

OFFER LETTER

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HARMAN

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4 December 2024



Adithya Awati
Bangalore, Karnataka

Sub: Offer of Internship

Hello Adithya,

Congratulations! Pursuant to our discussion, we are delighted to offer you employment opportunity with **Harman Connected Services Corp. India Pvt. Ltd., Bangalore**. Your initial place of posting will be **Harman India** situated at **IN_Bangalore_EOIZ Indust Area Campus_HCS, Bangalore, Karnataka**. In this capacity you will report to **Business Unit/Function Head** or his/her nominee.

Further, the "Company" for all purposes in this Offer of Internship ("Offer") shall mean and include the Affiliates of the Company, where the context may so require.

Following are the terms and conditions associated with your Internship:

1. Definitions:

"Company" means **Harman Connected Services Corp. India Pvt. Ltd.**

"Affiliate" shall mean, with respect to the Company, any individual, corporation, association or other Person (*defined as below*), which directly or indirectly, controls the Company, is controlled by the Company or is under common control with the Company. For this purpose, "control" (including, with its correlative meanings, "controlled by" and "under common control with") shall mean the possession, directly or indirectly, of the power to direct or cause the direction of management or policies of a Person, whether through the ownership of securities or partnership or other ownership interests, by contract or otherwise.

"Person" means an individual, company, association, joint venture, partnership, estate, trust, an association of persons and any other entity or organization, governmental or otherwise.

2. Date of Joining: Your Internship with the Company shall commence on **3 February 2025**. In the event you fail to report at the Company location mentioned above on the mentioned date, the offer made herein shall stand withdrawn, unless change in the reporting date is communicated and agreed with the Company in writing.

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Harman Connected Services Corporation India Pvt.Ltd.
CIN: U72200KA2002PTC030427 Email: contactus@harman.com

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

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We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Deepshree Buchade, Assistant Professor** – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Adithya N Awati [1DS21AI001]

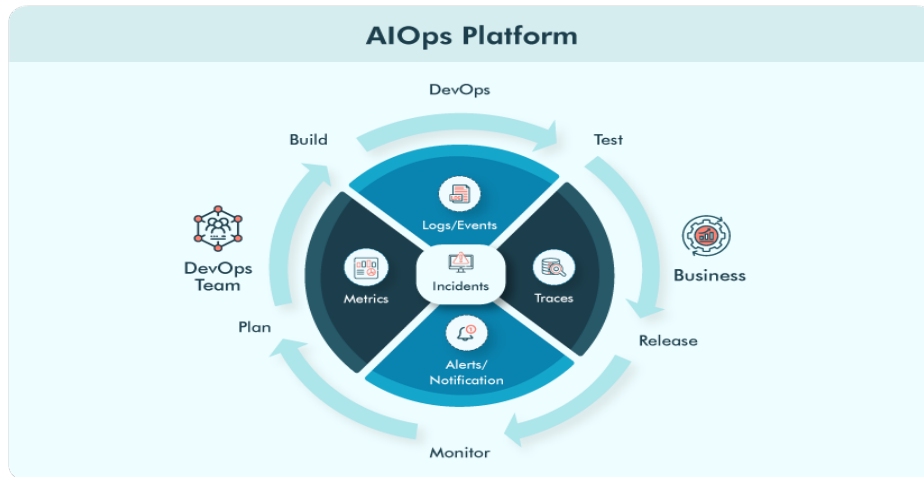
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CHAPTER 1

INTRODUCTION

1.1 Introduction to Data Science



Artificial Intelligence for IT Operations (AIOps) is an emerging discipline that applies artificial intelligence (AI) and machine learning (ML) techniques to enhance and automate various aspects of IT operations. Coined by Gartner in 2016, AIOps addresses the increasing complexity and data volume in modern IT environments by leveraging big data analytics to automate processes such as event correlation, anomaly detection, and causality determination.

In traditional IT operations, managing vast amounts of data generated by various components—such as servers, applications, and network devices—poses significant challenges. AIOps platforms ingest and aggregate this data, enabling intelligent analysis to identify significant events and patterns related to application performance and availability issues. By doing so, AIOps helps IT teams to detect and resolve issues more efficiently, often before they impact end-users.

In the context of cloud computing, AIOps plays a crucial role in managing dynamic and distributed systems. For instance, it enables dynamic scaling of resources based on real-time demand, ensuring optimal performance without overprovisioning. As organizations continue to adopt digital transformation initiatives, the role of AIOps becomes increasingly vital. By providing intelligent insights and automation capabilities, AIOps empowers IT teams to maintain high levels of service availability and performance in complex, hybrid environments.

1.2 Introduction to Generative AI



Generative AI (GenAI), a subfield of artificial intelligence, focuses on creating new content by learning patterns from existing data. Unlike traditional AI models that classify or predict based on input data, generative AI models are capable of producing novel data that mimics the characteristics of the training data. Prominent examples include Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and large language models like OpenAI's GPT (Generative Pretrained Transformer). These models can generate text, images, music, code, and even realistic voices. Applications range from creating chatbots and digital art to automating content creation and simulating complex environments in gaming and virtual reality.

The intersection of data science and generative AI opens up exciting opportunities for innovation. Generative AI can be used to create synthetic data for training machine learning models when real data is scarce or sensitive. It also aids in automating report generation, generating realistic simulations for testing, and improving data augmentation strategies. In turn, data science provides the foundational understanding and tools necessary to evaluate and refine generative models. As both fields continue to evolve, they are transforming industries by enabling smarter automation, deeper personalization, and more intuitive human-machine interactions. The combination of data science and generative AI is shaping the future of how we analyze, generate, and interact with data in an increasingly intelligent world.

1.4 CloudOps and Infrastructure

Cloud Operations, commonly referred to as CloudOps, is a comprehensive framework that encompasses the strategies, tools, and processes used to manage the delivery, performance, and optimization of IT services and workloads in cloud environments. It integrates elements of IT operations (ITOps), cloud computing, and DevOps principles to ensure continuous and efficient operations of cloud-based systems.

At its core, CloudOps focuses on automating software delivery, application management, and server management, all while provisioning resources from the cloud. This approach allows organizations to achieve high availability and scalability, as well as to optimize costs by dynamically adjusting resources based on demand.

Infrastructure plays a pivotal role in CloudOps, particularly through the use of Infrastructure as a Service (IaaS). IaaS provides virtualized computing resources over the internet, including servers, storage, and networking components. This model enables organizations to deploy and manage applications without the need to invest in or maintain physical hardware, offering flexibility and scalability to meet varying workload demands.

Effective CloudOps requires robust infrastructure management, which involves monitoring system performance, ensuring security, and maintaining compliance across different cloud environments. By leveraging automation and real-time analytics, CloudOps teams can proactively identify and resolve issues, thus minimizing downtime and enhancing the overall reliability of IT services.

In summary, CloudOps and its underlying infrastructure are essential for modern organizations aiming to deliver agile, scalable, and resilient IT services. By adopting CloudOps practices, businesses can better align their IT operations with strategic goals, drive innovation, and respond swiftly to changing market demands.

1.3 Agentic AI



Agentic AI refers to a class of artificial intelligence systems that are designed to operate as autonomous agents capable of making decisions, setting goals, and taking actions to achieve those goals with minimal or no human intervention. Unlike traditional AI systems that follow predefined instructions or respond to specific inputs, agentic AI possesses a higher degree of independence and adaptability. These systems can perceive their environment, reason about the best course of action, learn from experience, and act proactively. Examples include autonomous robots, personal AI assistants, and AI-powered agents in complex simulations or real-world scenarios like logistics, finance, and customer support.

What sets agentic AI apart is its ability to pursue long-term objectives, dynamically adjust plans, and collaborate with other agents or humans. This involves integrating multiple AI capabilities such as natural language processing, planning, reinforcement learning, and decision-making under uncertainty. With advancements in multi-agent systems and reinforcement learning, agentic AI is becoming more capable of functioning in complex, interactive environments. As the field matures, it raises important questions around ethics, control, and accountability, making responsible design and oversight essential to ensure these autonomous systems align with human values and societal goals.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Harman – American Electronics Company



2.1 Introduction

Harman International Industries, Inc., commonly known as **Harman**, is a leading American audio electronics company renowned for its innovative contributions to the audio and infotainment sectors. Headquartered in **Stamford, Connecticut**, Harman operates as a subsidiary of Samsung Electronics since its acquisition in **2017**. The company designs and engineers connected products and solutions for automakers, consumers, and enterprises worldwide, offering a diverse portfolio of brands including **JBL, Harman Kardon, AKG, and Infinity**.

The origins of Harman trace back to **1953** when **Sidney Harman** and **Bernard Kardon** founded Harman Kardon, introducing the world's first integrated hi-fi receiver. In 1956, Sidney Harman acquired full control of the company, leading to the establishment of Harman International. Over the decades, Harman expanded its footprint through strategic acquisitions, including notable audio companies such as JBL, Infinity, and AKG. The company also ventured into automotive audio, supplying premium sound systems to leading car manufacturers.

In the 1980s and 1990s, Harman continued its growth trajectory by acquiring companies like Lexicon and Becker, further solidifying its position in both consumer and professional audio markets. Recognizing the evolving technological landscape, Harman expanded into software and services, acquiring Symphony Teleca and Redbend in 2015 to enhance its capabilities in cloud, mobility, and analytics.

The acquisition by Samsung Electronics in 2017 marked a significant milestone, enabling Harman to leverage Samsung's global scale and resources to accelerate innovation in connected technologies. Today, Harman continues to lead in delivering cutting-edge audio and connected solutions across various industries, maintaining its legacy of excellence and innovation.

Harman International has established a significant presence in India since initiating operations in 2009. With a rapidly growing workforce exceeding 9,000 employees, the company operates from major development centers located in Bengaluru, Pune, Gurgaon, Noida, Chennai, Mumbai, and Coimbatore. These centers are integral to Harman's global operations, contributing to its Automotive, Lifestyle, and Digital Transformation Solutions businesses.

In alignment with India's 'Make in India' initiative, Harman has invested substantially in building its R&D and manufacturing capabilities within the country. A notable example is the state-of-the-art manufacturing facility and acoustics lab in Pune, which delivers cutting-edge infotainment and audio solutions for automotive partners, including TATA Motors and Maruti Suzuki.

Harman's commitment to innovation is further exemplified by the establishment of the HARMAN Experience Center in Bangalore. This multi-functional facility showcases the latest advancements in audio, lighting, video, and control systems, serving as a hub for customer engagement and technological demonstrations.

Beyond its commercial endeavors, Harman actively engages in community initiatives across India. The company supports various cultural and educational programs, such as A.R. Rahman's Sunshine Orchestra, Pune's Gaana Saraswati Mahotsava, and Bengaluru's Rangashankara theatre, reflecting its dedication to social responsibility and cultural enrichment.

Through these multifaceted activities, Harman International continues to strengthen its footprint in India, leveraging the country's talent and resources to drive innovation and deliver exceptional value to its global clientele.



2.2 Founders

Dr. Sidney Harman – Co-Founder



Dr. Sidney Harman, an American engineer and entrepreneur, **co-founded Harman Kardon** in 1953 alongside **Bernard Kardon**. Their collaboration led to the creation of the world's first integrated hi-fi receiver, marking a significant milestone in the audio industry. In 1956, Dr. Harman acquired full control of the company, establishing Harman International Industries. Beyond his contributions to audio technology, Dr. Harman was known for his progressive management philosophy, emphasizing employee empowerment and innovation. His leadership not only propelled Harman International to prominence in the audio electronics sector but also left a lasting impact on corporate culture and management practices.

Christian Sobottka – President and CEO



As of **April 1, 2025**, **Christian Sobottka** serves as the **President and Chief Executive Officer of Harman International**. Prior to this role, he led Harman's Automotive Division, where he played a pivotal role in advancing the company's connected car technologies and in-cabin experiences. With over 25 years in the automotive industry, Sobottka brings a wealth of experience to his leadership position. His appointment as CEO follows the tenure of Michael Mauser, who stepped down after nearly three decades with the company. Under Sobottka's leadership, Harman aims to continue its trajectory of innovation and market expansion, particularly in the realms of connected technologies for automotive, consumer, and enterprise markets.

2.3 Harman Connected Services – Harman India



Harman Connected Services (HCS) is a division of Harman International Industries, a wholly owned **subsidiary of Samsung Electronics**. HCS specializes in delivering comprehensive software engineering and digital transformation solutions to enterprises worldwide. The division offers a broad spectrum of services, including cloud engineering, cybersecurity, applied AI, data analytics, enterprise automation, and customer experience design. These services are tailored to meet the needs of various industries such as telecommunications, healthcare, industrial, and consumer sectors.

The origins of HCS trace back to the acquisition of **Symphony Teleca** by Harman International in **2015**. Symphony Teleca itself was formed through the merger of Symphony Services and Teleca, companies with a strong presence in software product engineering and mobile communications. Post-acquisition, Symphony Teleca was rebranded as Harman Connected Services, aligning with Harman's vision to expand its software services portfolio.

HCS has established a significant global footprint, with offices across the Americas, Europe, and Asia. **In India, HCS operates** in cities like **Bengaluru, Pune, Gurgaon, Chennai, and Mumbai**, contributing to Harman's largest employee base worldwide. The Indian division plays a crucial role in R&D and innovation, particularly in developing advanced infotainment and audio solutions for automotive partners.

Under the umbrella of Harman International, HCS continues to drive digital transformation initiatives, leveraging its deep domain expertise and technological capabilities to deliver intelligent, scalable, and impactful solutions to clients globally.

CHAPTER 3

SCOPE OF WORK

During my time with **Harman DTS** at **HCS** (Harman Connected Services), I had the opportunity to contribute to advanced initiatives at the intersection of **AIOps** and **CloudOps** within the **Genesis platform**—a scalable cloud-native platform designed for global digital transformation solutions for Software Development lifecycle (SDLC) governed by AI Agents. My role primarily involved developing intelligent automation capabilities to enhance infrastructure efficiency, reduce operational overhead, and improve system responsiveness through predictive scaling and smart observability mechanisms.

One of the core projects I undertook was the design and training of a **Proximal Policy Optimization** (PPO)-based Reinforcement Learning model aimed at automating **Kubernetes cluster scaling** operations. The model was trained using a custom observation space comprising a combination of system-level metrics fetched from the **Kubernetes API**, application-level metrics defined by internal **microservices**, **anomalous SLI** (Service Level Indicator) values detected using anomaly detection algorithms, and forecasted values of these **SLIs**. By incorporating these diverse data sources, the model could make **One-Look Ahead** (OLA) predictions, effectively generating scaling actions in anticipation of future workload spikes or degradations. This significantly improved the agility and reliability of our cluster management, aligning with our goals of autonomous infrastructure management under the AIOps initiative.

In parallel, I also worked on decoupling the **Genesis infrastructure** by modularizing the **Terraform configurations** and **state management**. This involved breaking down complex infrastructure scripts into reusable modules, introducing dynamic variable handling, and streamlining environment-specific deployments. To further operationalize this, I integrated the infrastructure automation with **GitHub Actions**, crafting custom composite actions that allowed team members to trigger standardized workflows for provisioning, updating, and tearing down cloud environments. These **reusable GitHub** actions were adopted across the **CloudOps team**, improving maintainability, reducing deployment times, and ensuring compliance with infrastructure governance policies.

My involvement wasn't limited to backend automation—I actively participated in cross-

functional collaboration during weekly stand-up meetings, where we brainstormed ideas, identified blockers, and shared updates across **engineering**, **DevOps**, and **data science** teams. Additionally, I prepared and presented monthly progress reports to the **HTS VP** and **GM**, which helped me refine my ability to communicate technical insights and demonstrate value at an executive level.

Through this experience, I also deepened my hands-on skills in Kubernetes, Helm, and various observability and telemetry tools including **OpenTelemetry**, **Prometheus**, **Grafana**, **Loki**, and **Tempo**. These tools helped us collect, visualize, and act upon distributed tracing data and performance metrics from microservices deployed in production environments. I gained a working understanding of inter-service communication protocols, context propagation in distributed systems, and microservices architecture best practices.

Moreover, I explored emerging AI frameworks such as **LangGraph** and **OpenAI Agents SDK**, evaluating their potential for future applications in self-healing systems and proactive incident management. I also developed a strong command over **Terraform** and **GitHub's** enterprise workflow configurations, which are now foundational to modern **DevOps** and **CloudOps** practices.

Overall, this internship provided me not only with technical experience in state-of-the-art tools and systems but also exposed me to real-world challenges in building scalable, resilient, and intelligent infrastructure. It has laid a solid foundation for a future career in AIOps and cloud engineering, equipping me with both the hands-on capabilities and the strategic mindset required to thrive in this space.

Overall, my work at the **DTS** division at **HCS** gave me the opportunity to engage with state-of-the-art technologies in **CloudOps**, **Agentic AI**, and **AIOps**, contributing to the development of smarter, more automated systems that enhanced both internal operations and customer experiences. The ability to blend cutting-edge AI techniques with data science and visualization tools allowed me to contribute to Harman's journey toward becoming a more agile, data-driven organization

CHAPTER 4

LEARNING OBJECTIVES

1. Understanding the Genesis Platform and Cloud Ecosystem:

During the internship, one of my foundational learning objectives was to gain a deep understanding of the Genesis platform—Harman’s scalable cloud-native framework that powers digital transformation initiatives. I explored its modular architecture and how it integrates with various tools, microservices, and infrastructure automation pipelines. This involved studying the design patterns behind service orchestration, data flow, and multi-cloud deployment models. I also developed an understanding of how Genesis interacts with CI/CD systems, observability stacks, and policy-driven resource provisioning. This ecosystem-level knowledge helped me grasp the full life cycle of cloud-native applications and understand how large-scale, production-grade platforms operate in dynamic, real-world conditions.

2. Proficiency in Public Cloud Platforms – AWS and Azure:

A major focus of my learning journey was to become proficient with the cloud service stacks used by the CloudOps team—primarily AWS and Microsoft Azure. I worked hands-on with core services like EC2, S3, IAM, VPCs, ECS, and EKS in AWS, and Azure equivalents such as Azure VMs, Blob Storage, Azure Kubernetes Service (AKS), and Azure Pipelines. I gained experience in provisioning cloud infrastructure, setting up secure access, and configuring network and storage resources, which helped me understand how different cloud providers implement similar services and the strategic choices involved in adopting multi-cloud strategies. I also explored cost optimization, scalability concerns, and disaster recovery principles in both environments.

3. Mastery of Kubernetes, Helm, and Observability Tools:

Another key learning objective was to build expertise in Kubernetes and its ecosystem, particularly Helm for application packaging and deployment. I gained practical experience deploying microservices on Kubernetes clusters, writing Helm charts, and managing version-controlled configurations. In tandem, I dove deep into observability practices using OpenTelemetry for telemetry data collection, Prometheus for monitoring, Grafana for visualization, and Loki and Tempo for log aggregation and distributed tracing. This experience helped me understand how to instrument systems for full-stack observability, troubleshoot performance issues, and correlate metrics, logs, and traces for end-to-end insight into system behavior.

4. Understanding Microservices Architecture and Distributed Systems:

My internship provided a real-world context to learn about distributed systems, inter-service communication, and context propagation. I explored service meshes, RESTful APIs, and gRPC, as well as mechanisms for correlation of service calls using trace IDs. I learned how to analyze distributed traces to understand dependencies between services, identify bottlenecks, and optimize throughput. I also studied various microservices design patterns such as circuit breakers, service discovery, and retries, which are essential for building resilient systems. These skills are critical in cloud-native application development and operations, especially in AIOps environments where system dynamics are complex and fast-changing.

5. Exploration of Reinforcement Learning for AIOps:

A particularly innovative part of my internship was exploring the application of reinforcement learning (RL) to AIOps. I implemented and trained a Proximal Policy Optimization (PPO) model to automate Kubernetes cluster scaling decisions. The model's observation space was designed to include not only system metrics but also anomaly-detected and forecasted SLIs, making it truly predictive. I studied the theoretical foundations of RL and understood how concepts like policy gradients, reward engineering, and exploration-exploitation trade-offs translate into real-world operational automation. This learning helped me see how AI can transform cloud infrastructure from reactive to proactive.

6. Familiarization with Agentic AI Frameworks:

I explored cutting-edge agentic AI frameworks such as LangGraph and the OpenAI Agents SDK. These tools enabled me to understand how autonomous agents can be designed to make decisions, execute tasks, and adapt to evolving environments within cloud operations. I experimented with building small agent workflows and learned how these can eventually be integrated into self-healing systems, automated incident triaging, and intelligent control loops. This was an exciting preview into the next wave of AI-infused DevOps.

7. Infrastructure as Code and GitOps Workflows:

Lastly, I gained extensive hands-on experience in Infrastructure as Code (IaC) using Terraform. I learned to define infrastructure components declaratively, manage state files, and structure configurations for reusability. I also integrated Terraform scripts with GitHub Actions to enable automated provisioning and de-provisioning workflows as part of a GitOps model. Additionally, I built reusable GitHub composite actions that streamlined pipeline development for the team. I also understood GitHub organization-level settings, secrets management, and role-based permissions, giving me a complete picture of managing cloud infrastructure through version-controlled workflows.

CHAPTER 5

CHALLENGES AND SOLUTION

1. Complexity of Observation Space

Challenge:

Integrating a diverse set of metrics—ranging from Kubernetes API server metrics, application-level metrics from internal microservices, to anomalous and forecasted SLIs—into a meaningful observation space posed a significant challenge. It required an in-depth understanding of how different components in a cloud-native microservice architecture behave and interact. Each metric had to be normalized, structured, and contextually aligned for the Reinforcement Learning (RL) agent to interpret the system's state effectively. This was particularly difficult given the high dimensionality and varying temporal behaviors of the metrics.

Solution:

To address this, I trained a Proximal Policy Optimization (PPO) based RL algorithm that could process this complex state space and generate One-Look Ahead (OLA) scaling actions. The model was built with an observation space that included live data from Kubernetes clusters, application metrics, anomalous SLIs (detected using anomaly detection algorithms), and forecasted SLI values. This approach enabled the RL agent to learn predictive auto-scaling policies. I also set up a real-time data streaming architecture using Kafka and Faust (a stream processing framework) to feed metrics into the model for inference. For training purposes, metrics were stored in TimescaleDB, enabling time-series queries and historical analysis for model training and evaluation.

2. Reward Design for RL in Cloud Environments

Challenge:

Designing an effective reward function for reinforcement learning in dynamic and unpredictable cloud workloads was particularly complex. The reward mechanism had to strike a balance between multiple conflicting objectives—minimizing resource costs, avoiding latency or downtime, and maximizing system responsiveness. In cloud environments where traffic patterns and resource utilization fluctuate, this becomes a non-trivial multi-objective optimization problem.

Solution:

I iteratively refined the reward function by incorporating performance indicators such as response time violations, CPU/memory utilization thresholds, and cost efficiency metrics. The model was trained to penalize both under-scaling (leading to poor performance) and over-scaling (leading to resource wastage), thereby learning a cost-performance tradeoff. The use of OLA prediction allowed the model to proactively allocate resources before an actual spike, significantly improving cluster stability and responsiveness.

3. Infrastructure Decoupling & Workflow Integration

Challenge:

The Genesis platform infrastructure was initially tightly coupled and complex to manage. Breaking down the Terraform configurations, maintaining consistent state across environments, and linking them dynamically with GitHub workflows required careful planning and robust implementation. Managing access, secrets, and backend state securely while ensuring modularity was also a major hurdle.

Solution:

I modularized the Terraform configurations into manageable modules and sub-modules, ensuring high cohesion and low coupling across different infrastructure layers. I then set up dynamic integration with GitHub workflows, where each module could be triggered independently based on the environment or change set. I implemented custom GitHub composite actions to simplify and standardize workflows for other team members. To secure and streamline state management, I configured OIDC-based identity providers with AWS and HashiCorp Cloud Platform (HCP) for GitHub authentication. This enabled smoother automation with robust, secure, and consistent Terraform state handling, making infrastructure-as-code practices more reliable and scalable.

4. Tooling and Ecosystem Mastery

Challenge:

Working on real-world cloud systems with tools like Kubernetes, AWS, Azure, Helm, GitHub Actions, and various observability stacks demanded rapid learning and practical application. With a steep learning curve and tight project timelines, understanding each tool's role and integrating them effectively into daily workflows was challenging.

Solution:

I approached this by dedicating time to hands-on experimentation and documentation-based learning. I deployed multiple microservices using Kubernetes and Helm charts, monitored them using Prometheus and Grafana, and configured logs and traces using Loki and Tempo. Through this, I built end-to-end observability pipelines. Simultaneously, I gained proficiency in GitHub Actions by developing complex CI/CD workflows, reusable actions, and automated deployment strategies. This hands-on exposure solidified my understanding and enabled me to contribute effectively to production-grade cloud environments.

5. Distributed Systems Debugging

Challenge:

Debugging microservices in a distributed environment required understanding complex communication protocols, context propagation, and service dependencies. Identifying bottlenecks or failures across services was difficult without centralized tracing and telemetry.

Solution:

To solve this, I set up distributed tracing using OpenTelemetry integrated with Grafana Tempo. I instrumented services to propagate trace context across HTTP/gRPC boundaries, allowing full visibility into inter-service requests. Using this data, I could visualize end-to-end request flows in Grafana, identify latency sources, and optimize service communication. This experience helped me gain confidence in managing observability in distributed systems and highlighted the critical role of tracing in cloud-native debugging.

HARDWARE AND SOFTWARE REQUIREMENTS

1. Hardware Requirements

- **Laptop/Workstation**
 - Minimum 16 GB RAM
 - Quad-core CPU or higher
 - SSD storage (at least 256 GB)
 - Network interface for secure VPN access
- **Cloud Infrastructure Access**
 - AWS Account with access to:
 - EC2, S3, IAM, VPC, Lambda, ALB, CloudWatch
 - Azure Account with access to:
 - AKS, VMs, Blob Storage, Networking Services

2. Software Requirements

- **Operating System**
 - Linux (Ubuntu/Debian-based preferred) or macOS
- **Browsers**
 - Brave / Firefox for accessing cloud consoles and dashboards
- **Communication & Collaboration**
 - Microsoft Teams / Zoom
 - Outlook
 - Confluence / Jira (for documentation and task tracking)
- **Version Control**
 - Git
 - GitHub (Enterprise)

3. Development Tools

- Code Editors/IDEs
 - Visual Studio Code (with extensions for Python, YAML, Terraform)
 - PyCharm (for Python-based development)
- Terminal Tools
 - KDE / GNOME Terminal
 - Zsh/Bash shell with Oh My Zsh
- Containerization
 - Docker
 - Docker Compose

4. Technologies & Platforms

- **Cloud Platforms**
 - **Amazon Web Services (AWS)**
 - EC2, S3, Lambda, IAM, VPC, ALB, CloudWatch, ACM
 - **Microsoft Azure**
 - AKS, VMs, Azure Pipelines
- **Container Orchestration**
 - Kubernetes (EKS/AKS)
 - Helm (for package management)
- **Infrastructure as Code**
 - Terraform
 - HCP Terraform (Remote Backend)
- **CI/CD**
 - GitHub Actions
 - GitHub Composite Actions
- **Streaming & Data Handling**
 - Kafka (Apache)
 - Faust (Python stream processing)
 - TimescaleDB (time-series PostgreSQL extension)

5. Observability & Monitoring Tools

- OpenTelemetry
- Prometheus
- Grafana
- Loki (for logging)
- Tempo / Jaeger (for distributed tracing)

6. AI/ML & AIOps Technologies

- **Reinforcement Learning**
 - Stable Baselines3, Huggingface
- **Anomaly Detection Algorithms**
 - PyOD
 - SKTime
- **Agentic Frameworks**
 - LangGraph
 - OpenAI Agents SDK
- **Python-based ML libraries**
 - PyTorch / TensorFlow (if used)
 - Scikit-learn
 - Pandas, NumPy, Matplotlib

7. Languages & Frameworks

- **Programming Languages**
 - Python (primary language for development and modeling)
 - YAML (for Helm, GitHub workflows)
 - HCL (Terraform Configuration Language)
 - Bash/Shell scripting
- **Web/Server Framework**
 - FastAPI (for backend services deployed on Lambda)

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

During my time at Harman Digital Transformation Solutions (DTS), under the Harman Connected Services (HCS) division, I have been fortunate to contribute meaningfully to several projects and initiatives at the intersection of AIOps, cloud-native infrastructure, and DevOps automation. My internship focused primarily on building intelligent automation frameworks and scalable infrastructure for the Genesis platform, a foundational system used internally within Harman CloudOps teams. Through hands-on work, I achieved a set of significant milestones that reflect both the technical challenges encountered and the knowledge gained while navigating them.

One of the most impactful contributions I made was the development of a Reinforcement Learning (RL)-based predictive auto-scaling engine for Kubernetes workloads. This project aimed to move beyond reactive, threshold-based scaling by enabling the system to anticipate future workload demands and scale resources proactively. The model I built used the Proximal Policy Optimization (PPO) algorithm—a robust and widely adopted method for policy-based reinforcement learning. The complexity of this solution stemmed from the rich and multi-dimensional observation space I constructed, which included real-time Kubernetes metrics (such as CPU, memory, and pod utilization), internal microservice-level metrics, anomalous Service Level Indicators (SLIs) identified using anomaly detection algorithms, and forecasted SLI values for short-term future intervals.

To enable the real-time prediction capability of the model, I set up a Kafka server to stream time-series metrics, which were consumed and processed using Faust, a stream-processing framework written in Python. This architecture allowed the RL model to operate continuously, ingesting live data and issuing predictive scaling actions—referred to as One-Look Ahead (OLA) actions—that could preemptively resize cluster resources before actual load spikes occurred. In parallel, I used TimescaleDB, a time-series optimized PostgreSQL extension, to store and query historical metrics during the model training and validation phase. This full-stack solution combined intelligent decision-making with scalable, cloud-native systems and illustrated a production-grade application of machine learning in operations (AIOps).

In addition to work on AIOps, I made substantial contributions to the CloudOps infrastructure side, particularly involving the Genesis platform's Terraform-based infrastructure-as-code setup. When I joined, the infrastructure was monolithic and difficult to manage in modular, environment-specific contexts. Recognizing this bottleneck, I took the initiative to refactor and decouple the Terraform configurations into reusable modules and sub-modules. This approach significantly improved the maintainability, readability, and reusability of the infrastructure code. I introduced better state management practices using dynamic backends powered by HashiCorp Cloud Platform (HCP) and integrated GitHub OIDC (OpenID Connect) authentication for secure, automated access control.

To further improve team efficiency, I developed reusable GitHub composite actions tailored for common CloudOps workflows, such as Terraform plan/apply pipelines, environment bootstrapping, and cloud credential management. These actions were built with modularity in mind and shared within the team for broader adoption. This contribution reduced pipeline complexity and lowered the entry barrier for new team members interacting with GitHub Actions for infrastructure management.

As a demonstration of my understanding of the Genesis ecosystem and AWS infrastructure stack, I led a mock deployment initiative in which I recreated the Genesis platform using a simplified yet realistic setup. I deployed a FastAPI-based web application on AWS Lambda, fronted by an Application Load Balancer (ALB), with HTTPS support via ACM-managed TLS certificates. The deployment included a secure VPC, subnet configurations, NAT gateways, routing tables, IAM roles and policies, and S3-based asset storage. The domain name was managed using OVH cloud's external DNS and validated through ACM for SSL provisioning. This comprehensive infrastructure was managed entirely through a new serverless Terraform module that I developed, giving me hands-on experience with critical AWS services such as Lambda, ALB, VPC, ACM, IAM, and CloudWatch. This project also served as a real-world testbed for applying infrastructure-as-code principles in a scalable, production-simulated environment.

Throughout the internship, I was actively involved in team routines such as weekly stand-up meetings, cross-functional brainstorming sessions, and monthly presentations to senior leadership, including the Vice President and General Manager of Harman Technology

Solutions (HTS). These forums provided an opportunity to share progress, receive feedback, and gain visibility into organizational goals and long-term strategies. Presenting technical work in a business-aligned manner helped sharpen my communication and storytelling abilities—an essential skill for bridging the gap between engineering and leadership.

Apart from technical contributions, this internship facilitated deep learning and skill development in a variety of tools, platforms, and methodologies. I became proficient in working with AWS and Azure cloud platforms, understanding how services such as EKS, AKS, Lambda, IAM, VPC, and storage layers are structured and secured. I gained extensive hands-on experience with Kubernetes, Helm, and observability tools including Prometheus, Grafana, Loki, and Tempo. Setting up and configuring OpenTelemetry for distributed tracing enhanced my understanding of how context propagates through microservices and how to effectively debug and monitor large-scale distributed systems.

On the machine learning side, I expanded my understanding of reinforcement learning in cloud contexts, with a focus on model tuning, reward engineering, and real-time inference pipelines. I also explored emerging technologies such as LangGraph and the OpenAI Agents SDK, which are used to create agentic AI frameworks that may play a future role in adaptive cloud systems. On the DevOps front, I developed skills in Terraform, GitHub Actions, OIDC-based cloud authentication, and secure automation practices.

In summary, the milestones I achieved during my internship reflect a strong alignment between my learning objectives and the real-world problems I was exposed to. I was able to apply and extend my knowledge across multiple domains: artificial intelligence in operations (AIOps), cloud-native application deployment, observability, infrastructure automation, and modern DevOps tooling. More importantly, the work I did had a tangible impact on the Genesis platform's operational maturity, performance, and maintainability. These experiences have given me not only technical depth but also a practical perspective on solving production-level challenges in fast-paced, high-scale engineering environments. The guidance, collaboration, and support from the Harman DTS team have made this journey both intellectually fulfilling and professionally valuable.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship at **Harman Connected Services (HCS)**, under the Digital Transformation Solutions (DTS) pillar, has been a transformative experience—both technically and professionally. Being embedded within a cutting-edge **CloudOps** and **AIOps** environment allowed me to engage with real-world challenges, apply my academic knowledge, and evolve as an engineer capable of contributing to large-scale enterprise systems.

From the outset, I was given the opportunity to work on impactful problems, particularly around predictive auto-scaling and infrastructure modularization in the Genesis platform. These weren't merely sandbox tasks but real responsibilities that demanded critical thinking, innovation, and deep system understanding. The level of trust and responsibility I was granted significantly enhanced my problem-solving skills and taught me how to take ownership of long-term, scalable solutions.

A standout element of my experience was the exposure to advanced **AIOps** techniques, specifically reinforcement learning applied to cloud resource scaling. Working on a **PPO**-based RL model taught me how AI can be used not just for analytics but for intelligent decision-making in real-time systems. I learned how to architect observation spaces from complex metric streams, design effective reward functions that balance performance and cost, and build real-time pipelines using **Kafka** and **Faust**. These insights extended far beyond theoretical machine learning and introduced me to the nuances of production-readiness and operational efficiency in intelligent systems.

Equally valuable was the hands-on experience I gained in cloud infrastructure, **DevOps** tools, and automation frameworks. Refactoring **Terraform** configurations into modular components and integrating them into secure GitHub Actions workflows taught me the importance of maintainable **infrastructure as code** (IaC). I was also able to design secure cloud authentication mechanisms using **OpenID Connect** and **HCP backends**—experiences that sharpened my skills in automation, identity management, and CI/CD.

The internship also emphasized the critical role of observability in distributed systems. Learning tools such as **Prometheus**, **Grafana**, **Loki**, and **Tempo** helped me understand the importance of **metrics**, **logs**, and **traces** in debugging and optimizing **microservices**. I developed a strong foundation in distributed tracing and context propagation, which are key to understanding system behavior across complex service meshes.

One of the most fulfilling aspects of my experience was the **culture** at **HCS**. Regular technical discussions, stand-up meetings, and brainstorming sessions encouraged collaboration, critical thinking, and continuous feedback. Presenting my work to senior leadership pushed me to communicate clearly, connect technical outcomes to business objectives, and defend my design choices. These interactions honed my communication, presentation, and stakeholder engagement skills—qualities that are equally important as technical proficiency.

On a personal level, the internship helped me grow in confidence and adaptability. Working across a wide array of tools, frameworks, and platforms—from **Kubernetes** to **Terraform**, from **PPO models** to **GitHub automation**—pushed me out of my comfort zone and gave me a more holistic understanding of modern cloud-native ecosystems. I now feel more equipped to pursue a career in **AIOps** and **Cloud Engineering**, with a deeper appreciation of the interdisciplinary skillset it demands—**spanning AI, DevOps**, and distributed system design.

Overall, my journey at HCS has been a rich learning experience, marked by hands-on engineering, strategic thinking, and meaningful collaboration. The work I contributed to not only helped solve real technical problems but also laid a strong foundation for my aspirations in building intelligent, scalable, and autonomous cloud platforms in the future.

CHAPTER 8

RECOMMENDATIONS

1. Hands-On Learning with Real-World Impact

At Harman, interns are entrusted with meaningful projects that directly contribute to the company's objectives. Rather than being limited to observational roles, I was actively involved in developing a reinforcement learning-based auto-scaling model for Kubernetes clusters. This project not only enhanced my technical skills but also provided a tangible sense of accomplishment as my work had a direct impact on the platform's efficiency. Such immersive experiences are a testament to Harman's commitment to providing interns with substantial learning opportunities.

2. Supportive and Collaborative Work Environment

The culture at Harman is characterized by its inclusivity and supportiveness. From day one, I felt welcomed and valued by my colleagues and mentors. Regular team meetings, open-door policies, and a culture of feedback fostered an environment where I could freely share ideas and seek guidance. This collaborative atmosphere not only facilitated my professional growth but also made my internship experience enjoyable and fulfilling.

3. Exposure to Cutting-Edge Technologies

Harman's commitment to innovation is evident in the technologies and tools utilized across projects. During my tenure, I had the opportunity to work with advanced tools and frameworks such as AWS, Azure, Kubernetes, Helm, OpenTelemetry, Prometheus, and Grafana. This exposure not only broadened my technical expertise but also prepared me to tackle complex challenges in modern cloud-native environments.

4. Emphasis on Professional Development

Beyond technical skills, Harman places a strong emphasis on the holistic development of its employees. I participated in various workshops and training sessions that enhanced my soft skills, including communication, leadership, and project management. These opportunities for growth are indicative of Harman's dedication to nurturing well-rounded professionals.

5. Global Presence with Local Impact

As a global leader in connected technologies, Harman offers the unique advantage of working on projects that have a worldwide impact while being rooted in local innovation hubs. My experience in the Bengaluru office allowed me to collaborate with diverse teams and understand global business dynamics, all while contributing to solutions tailored for local markets.

CHAPTER 9

CONCLUSION

My internship at Harman Connected Services (HCS) has been an immensely rewarding and transformative experience. It provided me with a unique opportunity to apply my academic foundations to real-world challenges at the intersection of Artificial Intelligence, Cloud Computing, and DevOps. From developing reinforcement learning-based auto-scaling models to modularizing infrastructure with Terraform and building secure, automated CI/CD pipelines, I was able to contribute to meaningful projects within the Genesis platform and the CloudOps team.

This journey has significantly deepened my technical expertise in Kubernetes, AWS, observability stacks, GitHub Actions, and infrastructure-as-code, while also sharpening my understanding of distributed systems, cloud-native architectures, and intelligent automation. Beyond the technical scope, I developed valuable soft skills through regular team collaboration, stakeholder presentations, and agile development practices.

The experience not only reinforced my interest in AIOps and Cloud Engineering but also equipped me with the confidence and skills required to pursue a career in building scalable, intelligent, and resilient systems. I am grateful to the mentors, teammates, and leaders at Harman for their guidance and support throughout this internship. As I move forward, I carry with me the technical learning, professional growth, and collaborative spirit fostered during my time at HCS.

CHAPTER 10

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